Chapter

Library Classes

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1. What is the output of the below code segment? **MAIN 2024**

String x = “I love programming”?
StringBuffer y = new StringBuffer(x);
StringBuffer z = y.reverse();
System.out.print (z);

(a) programming love I
(b) evol I programming
(c) gnimmargorp love I
(d) gnimmargorp evol I

2. Which of the following is the wrapper class for simple data type-char?
MAIN 2024

(a) String
(b) Char
(c) Character
(d) Float

3. Which of the following is the output of the below code snippet? COMP 2022

```
double x = -3.142;
int y = (int) Math.abs(x) + (int) x;
System.out.print(y);
```

(a) 0
(b) 6.284
(c) 6.0
(d) -6

4. Which of the following methods of Java returns the nearest and larger whole number which is greater than or equal to the argument? MAIN 2023

(a) Ceil
(b) Floor
(c) Trim
(d) Max

5. When an object of a class is converted to a simple data type, it is called ________ SQP 2021

(a) Unboxing
(b) Boxing
(c) Explicit type conversion
(d) Implicit type conversion

6. Can any primitive type be a class member variable? MAIN 2023

7. How can a class type variable support the accessing of members of a class? SQP 2022

8. Can another class be a member of a class? COMP 2021

9. What are library classes? Give an example. MAIN 2021

10. Write a Java statement to input/read the following from the user using the keyboard: COMP 2022

(i) Character
(ii) String

11. State the method that: MAIN 2023

(i) Converts a string to a primitive float data type.
(ii) Determines if the specified character is an uppercase character.

12. What is the data type that the following library functions return? COMP 2024

(i) isWhitespace(char ch)
(ii) Math.random()

13. Where are these classes grouped together and placed? COMP 2024

14. Write a program to design a class Box with members as follows :
Data Members : length, breadth and height**. SQP 2021**
Member Methods :
(i) Compute volume of the box.
(ii) Compute the total area of the box.

Solution

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1. (d) gnimmargorp evol I

- StringBuffer y = new StringBuffer(x); creates a StringBuffer object y with the content "I love programming".
- y.reverse(); reverses all characters in the buffer, producing "gnimmargorp evol I".
- System.out.print(z); prints the reversed string.

Hence, the output is **gnimmargorp evol I**.

2. (c) Character

In Java, the primitive type char has a corresponding wrapper class Character. Wrapper classes allow primitive data types to be treated as objects, enabling their use in collections, generics, and other object-oriented features.

3. (a) 0

Explanation:

- x = -3.142
- Math.abs(x) = 3.142 → cast to int → 3
- (int)x = -3
- Adding: 3 + (-3) = 0

Hence, the output is **0**.

. (a) Ceil

The Math.ceil() method in Java returns the smallest integer that is **greater than or equal** to the given argument. It effectively rounds **up** to the nearest whole number.

5. (a) Unboxing

Explanation:
Unboxing in Java is the process of converting an **object of a wrapper class** (like Integer, Double, Character) into its corresponding **primitive data type** (like int, double, char).

6. Yes, primitive types can be class member variables in Java. These variables store simple values such as integers, floating-point numbers, characters, or boolean values. Primitive type member variables are defined inside a class and can be accessed and manipulated using instances of that class.

7. Class members can be accessed through class type variables (objects) by using the dot (.) operator. e.g.,
object. memberFunction () ;

8. Yes. Another class can be a member of a class in Java using **nested (inner) classes**, which can access the private members of the enclosing class.

9. Library classes: Pre-defined classes in Java that are included in a program for use. They are also called standard or built-in Java classes and contain ready-made member variables, constructors, and methods.

Example: String, Math, Scanner.

10. (i) char x = (char) buf.read(); // buf is a BufferedReader object
(ii) String y = buf.readLine();

11. (i) parseFloat()
(ii) isUpperCase()

12. (i) Boolean
(ii) Double

13. These library classes are grouped together as **packages**. A collection of such packages forms the **Java Class Library**.

14.

```
import java.util.*;
public class Box {
    int l, b, h;

    void Accept() {
     Scanner buf = new Scanner(System.
in);
       System.out.println("Enter the
length, breadth, and height");
        l = buf.nextInt();
        b = buf.nextInt();
        h = buf.nextInt();
    }

    void volume() {
        int vol = l * b * h;
       System.out.println("VOLUME OF
THE BOX: " + vol);
    }

    void area() {
        int a = 2 * (l * b + b * h +
l * h);
         System.out.println("Area of
box: " + a);
    }

    public static void main() {
        Box ob = new Box();
        ob.Accept();
        ob.volume();
        ob.area();
    }
}
```